

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**

**Applied Business Analytics Practical**

**Practical – IX**

|                                |                                     |
|--------------------------------|-------------------------------------|
| Roll No.: T005                 | Name: Ritesh Chaubey                |
| Class: TYIT                    | Batch: 1                            |
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**Aim : Describing the Distribution of a Variable**

**Part A: Hypothesis Testing**

**Dataset: supermarket\_sales.csv**

- 1. Load the supermarket\_sales.csv dataset.**
- 2. Display the first five records.**
- 3. Select the revenue column.**
- 4. Check for missing values.**
- 5. Calculate the mean revenue.**
- 6. Assume that the expected average revenue is ₹1,500.**
- 7. Write the null hypothesis:**  
 **$H_0: \mu = 1500$**
- 8. Write the alternative hypothesis:**  
 **$H_1: \mu \neq 1500$**
- 9. Set the significance level as 5%.**
- 10. Perform a one-sample t-test.**
- 11. Calculate the p-value.**
- 12. Compare the p-value with 0.05.**
- 13. State whether the null hypothesis is rejected or not rejected.**
- 14. Write the conclusion in simple business terms.**

**CODE :-**

```
import pandas as pd
from scipy import stats
# Load dataset
df = pd.read_csv("supermarket_sales(2).csv")
# Display first 5 records
print("First 5 Records:")
print(df.head())
# Select revenue column
revenue = df["revenue"]
# Check missing values
print("\nMissing Values:")
print(revenue.isnull().sum())
```

```
# Remove missing values
revenue = revenue.dropna()
# Calculate mean
mean_revenue = revenue.mean()
print("\nMean Revenue:", mean_revenue)
# Hypotheses
print("\nH0: Mean Revenue = 1500")
print("H1: Mean Revenue != 1500")
# Significance level
alpha = 0.05
# One-sample t-test
t_stat, p_value = stats.ttest_1samp(revenue, 1500)
# Display results
print("\nT-statistic:", t_stat)
print("P-value:", p_value)
# Decision
if p_value < alpha:
    print("\nDecision: Reject H0")
    print("Conclusion: Mean revenue is significantly different from 1500.")
else:
    print("\nDecision: Do not reject H0")
    print("Conclusion: There is not enough evidence that mean revenue is different from 1500.")
```

```
First 5 Records:
  revenue
0    1450
1    1520
2    1380
3    1600
4    1490

Missing Values:
0

Mean Revenue: 1501.1666666666667

H0: Mean Revenue = 1500
H1: Mean Revenue != 1500

T-statistic: 0.09017263867681004
P-value: 0.9287699561807621

Decision: Do not reject H0
Conclusion: There is not enough evidence that mean revenue is different from 1500.
```

## Part B: A/B Testing

**Dataset:** ab\_testing.csv

**The dataset contains:**

**Group Users Conversions**

**A 1000 120**

**B 1000 150**

- 1. Load the A/B testing dataset.**
- 2. Display the data.**
- 3. Calculate the conversion rate for Group A.**
- 4. Calculate the conversion rate for Group B.**
- 5. Compare the conversion rates of A and B.**
- 6. Write the null hypothesis:**  
**H0:  $p_A = p_B$**
- 7. Write the alternative hypothesis:**  
**H1:  $p_A \neq p_B$**
- 8. Set the significance level as 5%.**
- 9. Perform a two-proportion test.**
- 10. Calculate the p-value.**
- 11. Compare the p-value with 0.05.**
- 12. State whether the null hypothesis is rejected or not rejected.**
- 13. Create a bar chart to compare the conversion rates.**
- 14. Write the conclusion based on the test result.**

**Code:-**

```
import pandas as pd
from scipy import stats
from statsmodels.stats.proportion import proportions_ztest
import matplotlib.pyplot as plt
# Load dataset
df = pd.read_csv("ab_testing.csv")
# Display dataset
print("A/B Testing Dataset:")
print(df)
# Get values for Group A
users_A = df.loc[df["Group"] == "A", "Users"].iloc[0]
conversions_A = df.loc[df["Group"] == "A", "Conversions"].iloc[0]
# Get values for Group B
users_B = df.loc[df["Group"] == "B", "Users"].iloc[0]
conversions_B = df.loc[df["Group"] == "B", "Conversions"].iloc[0]
# Calculate conversion rates
rate_A = conversions_A / users_A
rate_B = conversions_B / users_B
print("\nGroup A Conversion Rate:", rate_A * 100, "%")
print("Group B Conversion Rate:", rate_B * 100, "%")
# Hypotheses
print("\nH0: Conversion Rate A = Conversion Rate B")
```

```
print("H1: Conversion Rate A != Conversion Rate B")
# Significance level
alpha = 0.05
# Two-proportion z-test
conversions = [conversions_A, conversions_B]
users = [users_A, users_B]
z_stat, p_value = proportions_ztest(
    conversions,
    users,
    alternative="two-sided"
)
print("\nZ-statistic:", z_stat)
print("P-value:", p_value)
# Decision
if p_value < alpha:
    print("\nDecision: Reject H0")
    print("Conclusion: There is a significant difference between Group A and Group B.")
else:
    print("\nDecision: Do not reject H0")
    print("Conclusion: There is no significant difference between Group A and Group B.")
# Scatter plot
plt.scatter(
    df["Users"],
    df["Conversions"]
)
plt.xlabel("Users")
plt.ylabel("Conversions")
plt.title("Users vs Conversions")
plt.show()
# Bar chart for conversion rates
plt.bar(
    ["Group A", "Group B"],
    [rate_A * 100, rate_B * 100]
)
plt.xlabel("Group")
plt.ylabel("Conversion Rate (%)")
plt.title("A/B Testing Conversion Rates")
plt.show()
```

**Output:-**

```
A/B Testing Dataset:
  Group Users Conversions
0    A   1000        120
1    B   1000        150

Group A Conversion Rate: 12.0 %
Group B Conversion Rate: 15.0 %

H0: Conversion Rate A = Conversion Rate B
H1: Conversion Rate A != Conversion Rate B

Z-statistic: -1.963049807622355
P-value: 0.04964038686536878

Decision: Reject H0
Conclusion: There is a significant difference between Group A and Group B.
```

